Epilepsy-relevant injury, energy, and signaling programs across mouse phases

Frozen Bridge Hallmark head

Acute Intermediate Chronic

Seizure injury / inflammation
- Unfolded protein response
- p53 Pathway
- Interferon-α response
- Reactive oxygen species
- Apoptosis
- TNFα signaling via NF-κB
- Interferon-γ response
- IL-6 / JAK / STAT3 signaling
- Inflammatory Response
- Complement

Energy / redox metabolism
- Peroxisome
- Heme metabolism
- Cholesterol homeostasis
- Hypoxia
- Fatty-acid metabolism
- Oxidative phosphorylation
- Glycolysis

Growth / injury signaling
- KRAS signaling DN
- mTORC1 signaling
- TGF-β signaling
- KRAS signaling UP
- PI3K / AKT / mTOR signaling

−3 −2 −1 0 1 2 3

Mouse kainate − phase-matched control (Disease − matched-control score)
← lower in epilepsy higher in epilepsy →

Conventional Hallmark GSEA

Acute Intermediate Chronic

−2 −1 0 1 2

Mouse kainate − phase-matched control (Normalized enrichment score (NES))
← lower in epilepsy higher in epilepsy →

filled: supported open: unresolved Seizure injury / inflammation Energy / redox metabolism Growth / injury signaling

Within each prespecified biological block, pathways are ordered by the widest Bridge phase swing; GSEA uses the identical row order. Bridge support: biological-sample bootstrap interval excludes zero; GSEA support: FDR q<0.05.